SENIOR HIGH SCHOOL • COMPUTER PROGRAMMING (.NET TECHNOLOGY)

Z-index, Position & Float

Lesson 7

Grade 11 | Quarter 2 | CPNT 2.5

LESSON

7

Z-index, Position & Float

Arranging and layering elements for real layouts

CPNT 2.5

What You Will Be Able To Do

Unpacked Performance Standards

1. Apply the CSS position property in arranging webpage elements.

2. Apply the CSS float property in creating side-by-side layouts.

3. Use the CSS z-index property to manage the stacking order of elements.

LESSON 7 • DISCUSSION

The CSS Position Property

The position property decides how an element is placed relative to the normal page flow.

`static (default)`

The normal flow — elements stack top to bottom. top/left/right/bottom have no effect.

`relative`

Nudges an element from its normal spot using top/left/etc., without affecting other elements.

`absolute`

Removes the element from flow and positions it relative to its nearest positioned ancestor.

`fixed & sticky`

fixed pins an element to the viewport (stays put on scroll); sticky switches to fixed only past a scroll point.

LESSON 7 • DISCUSSION

Float & Z-Index

Float creates side-by-side layouts; z-index controls what sits on top when elements overlap.

`float`

Pulls an element left or right, letting other content wrap around it — the classic side-by-side layout trick.

`Stacking Context`

z-index only works on positioned elements (not static) — each one can create its own stacking context.

`z-index`

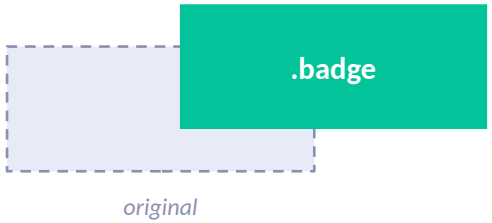
Controls stacking order when elements overlap — a higher number sits closer to the viewer.

LESSON 7 • EXAMPLES 1 & 2

Relative & Absolute Positioning

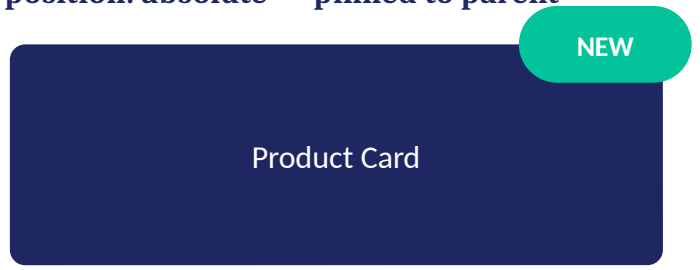
relative nudges an element from its own spot; absolute pulls it out of flow entirely.

1. position: relative — nudges from its spot




```
.badge {
  position: relative;
  top: -20px; left: 30px;
}
```

2. position: absolute — pinned to parent



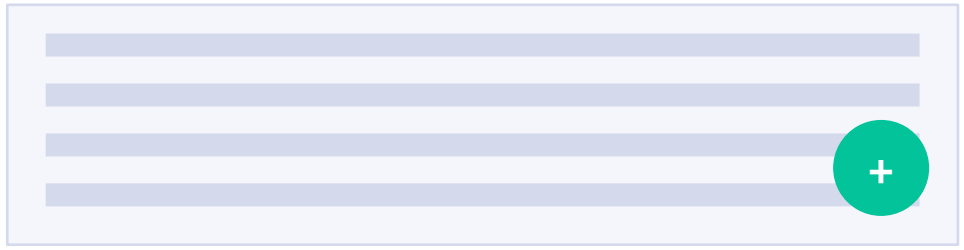

```
.card { position: relative; }
.badge {
  position: absolute;
  top: -8px; right: -8px;
}
```

LESSON 7 • EXAMPLES 3 & 4

Fixed & Sticky Positioning

fixed stays glued to the viewport; sticky toggles between relative and fixed as the page scrolls.

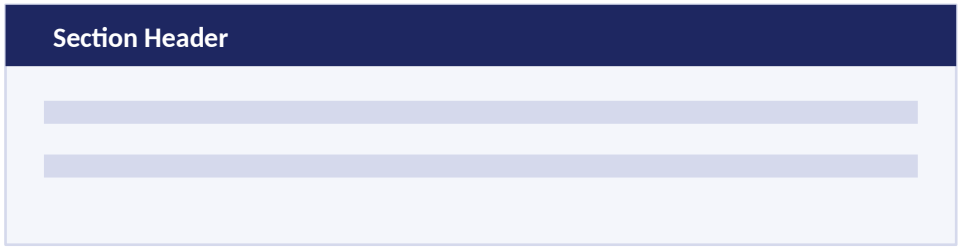
3. position: fixed — floating button



stays in place while scrolling

```
.fab {
  position: fixed;
  bottom: 24px; right: 24px;
}
```

4. position: sticky — header sticks on scroll



scrolls with content, then sticks

```
.section-header {
  position: sticky;
  top: 0;
}
```

LESSON 7 • EXAMPLES 5 & 6

Float Layouts & the Clearfix

float builds side-by-side columns; clearfix keeps their parent container from collapsing.

5. float — two-column layout




```
.sidebar { float: left; width: 30%; }
.main { float: left; width: 70%; }
```

6. clearfix — fixing a collapsed container




```
.container::after {
  content: ""; display: table;
  clear: both;
}
```

LESSON 7 • EXAMPLES 7 & 8

Stacking Order in Practice

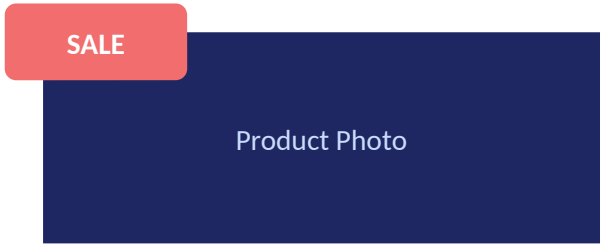
z-index decides who's on top; combined with position, it powers real UI patterns like image badges.

7. z-index — who's on top?




```
.back { position: relative; z-index: 1; }
.front { position: relative; z-index: 2; }
```

8. Real pattern: badge over a thumbnail




```
.thumbnail { position: relative; }
.badge {
  position: absolute; top: -8px; left: -8px;
  z-index: 2;
}
```

LESSON 7 • ASSESSMENT

Quick Check: Position, Float & Z-Index

Formative assessment — short quiz (identification / short answer), answered individually before the layout activity.

1. What is the default value of the position property, and what does it do?
2. What is the key difference between position: relative and position: absolute?
3. Which position value keeps an element fixed to the viewport even while scrolling?
4. What CSS technique is commonly used to prevent a container from collapsing around floated children?
5. If two positioned elements overlap, which one appears on top — the one with the higher or lower z-index?

LESSON 7 • ACTIVITY

Hands-On: Arrange & Layer Elements

Layout Activity: Positioned Badge & Two-Column Layout

Apply position, float, and z-index to arrange elements the way real websites do.

1. Build a two-column layout (sidebar + main content) using float.
2. Apply a clearfix to the container so it doesn't collapse around the floats.
3. Add a small badge or label positioned absolutely over a card or image.
4. Use z-index to make sure the badge always appears above the image.

NEXT UP

Lesson 8: CSS Image Gallery

With arranging and layering mastered, we'll put it to work building a responsive image gallery in Quarter 2's CPNT 2.6.

Thank you!